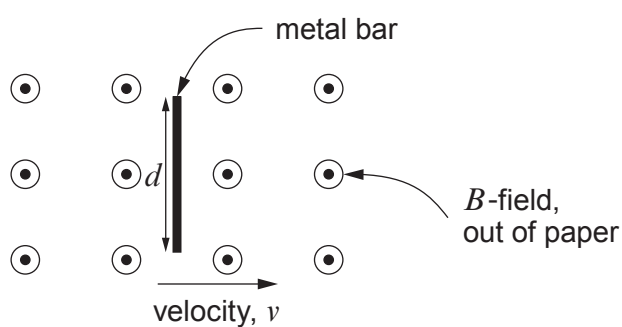


6. (a) The diagram shows a metal bar moving with velocity, v , through a uniform magnetic field, B .



Show that the rate of cutting of flux is Bvd .

[2]

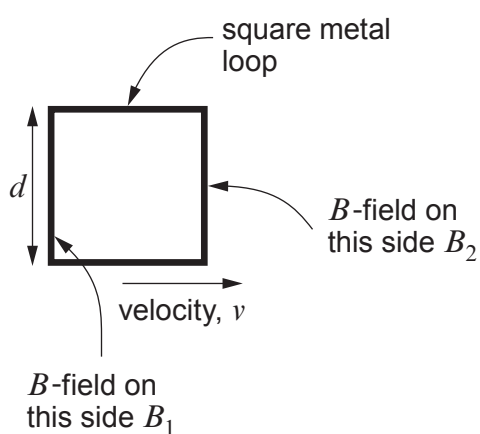
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- (b) A square metal loop moves through a **non-uniform** B -field with the field coming out of the paper.



- (i) Use the expression in part (a) to explain why the induced emf in the square metallic loop is given by:

[3]

$$(B_2 - B_1)vd$$

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- (ii) The B -field, through which the square metal loop moves, increases with distance at a rate of 1.20 T m^{-1} (in the direction of the velocity). The square loop has sides of length 3.8 cm and moves at a speed of 8.8 m s^{-1} . Show that the induced emf in the square metal loop is approximately 15 mV .

[2]

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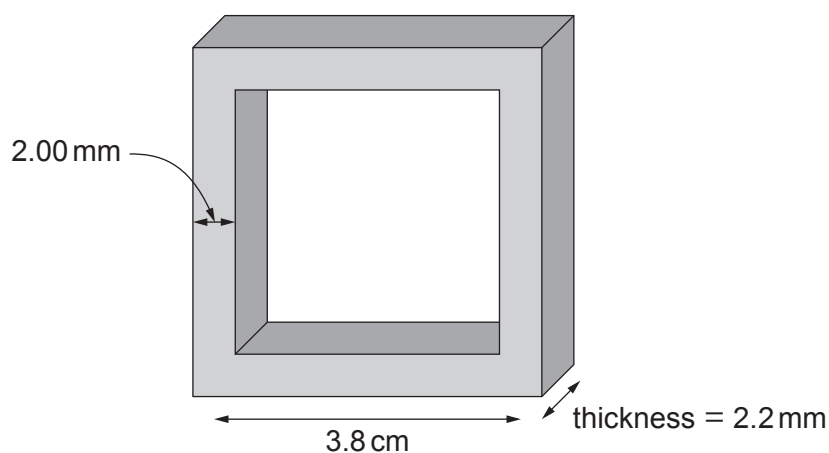
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- (iii) The square metal loop is made of silver of resistivity $1.59 \times 10^{-8} \Omega \text{m}$ and its dimensions are shown in the diagram below:



Calculate the current in the square loop.

[5]

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